Compressed Sensing

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Chapter 1

Compressed Sensing

Source: Lecture by Prof Michael Lusting, Berkeley University.

1.1 Introduction and Motivation

- Usually when we store information (images captured by camera, some audio files, $\cdots$), we store them in a compressed way, as shown in Figure 1.1

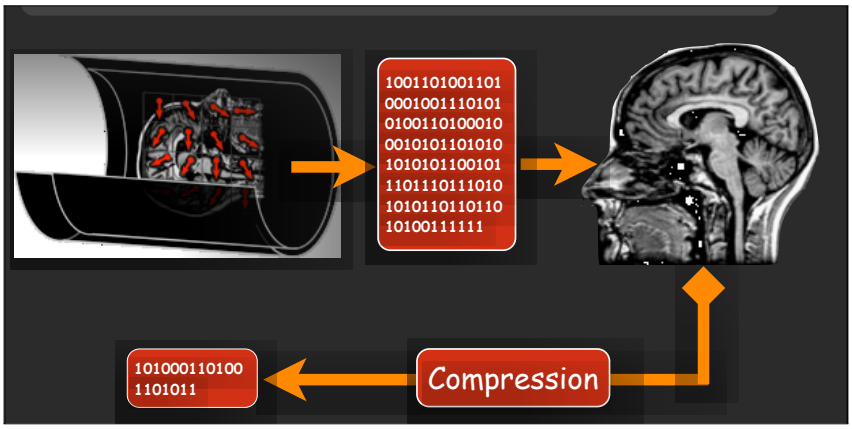


Figure 1.1: Compressing MRI: first we collect, then we compress

  - In other words, depending on Figure 1.1, we usually spend allot of time constructing the data (the measurements), construct the image (the MRI), then inputted to compression algorithm to compress it and ***throw away allot of measurements we spend time to collect them !***.

- As a side note for compression: we usually compress data depending on human perception (hearing or seeing), in such a way we restore only valuable information (so not all the data) and we can't see or hear (in case of audio) the difference

- But what if ***we can sense directly*** the information, as shown in Figure 1.1

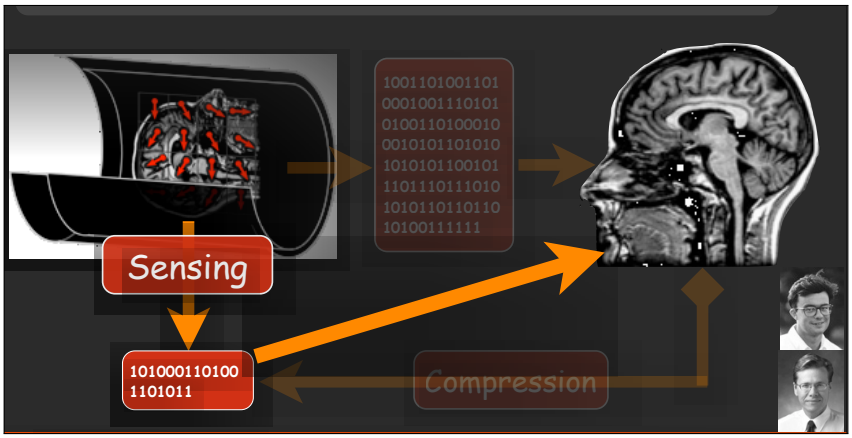


Figure 1.2: Sensing Concept

The approach shown in Figure 1.1 is very helpful because by this way, we don't need to throw allot of information we gathered as in Figure 1.1, and directly collect the valuable information needed for reconstruction.

1.2 Example of Compression and Interpretation

- First let's see some audio signal in Figure 1.3

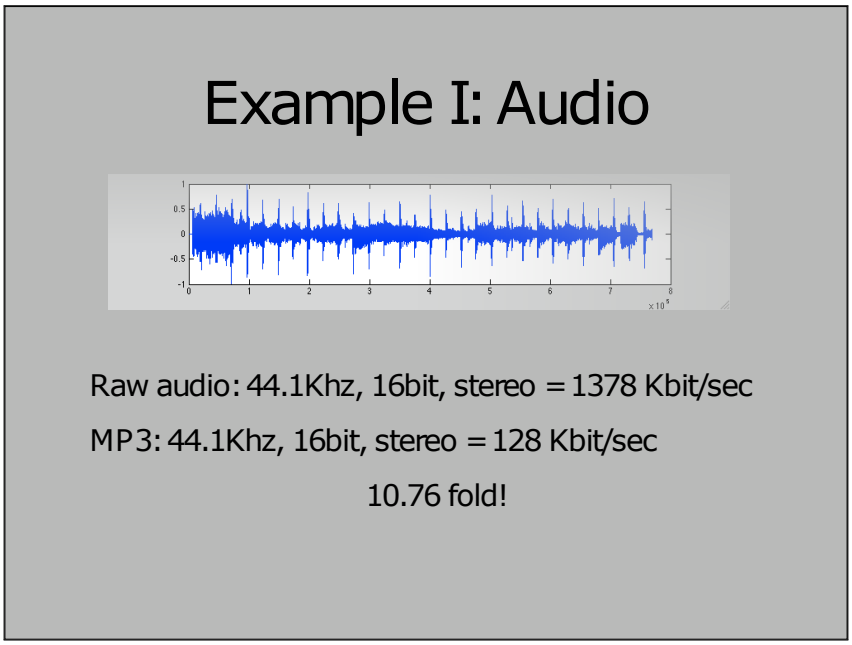


Figure 1.3: Audio Compression

  - As interpretation, if we listen at a rate of 128 Kbit/s, it would make no difference if we here in original rate

- What about images ? are images compressible like audio.

Images are generally more complex type of signals, because they are higher in dimension.

An example is shown in Figure 1.4.

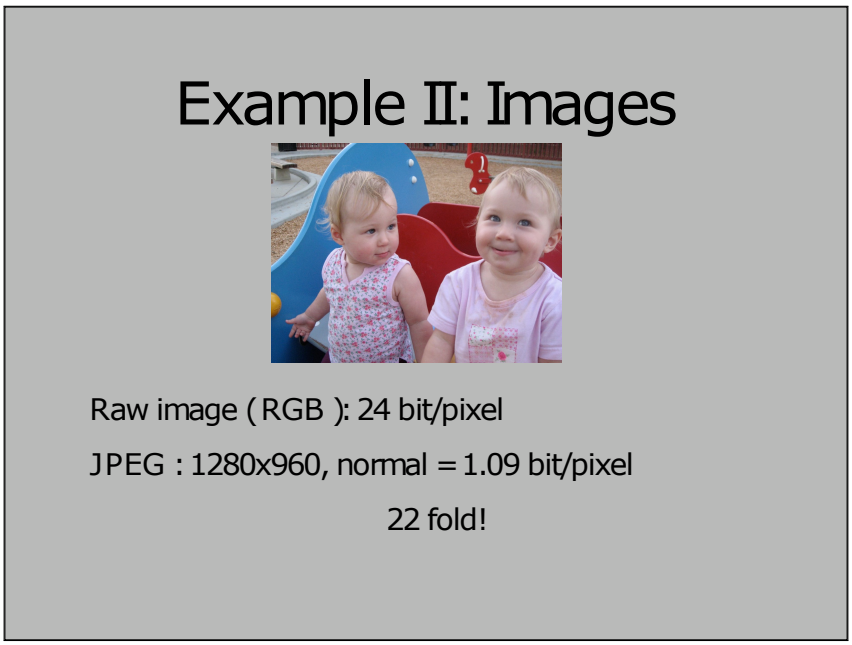


Figure 1.4: Image Compression

- It turns out that we can compress with higher compression ration
- It is true that dimension increase, but the information doesn't necessary increase also as the dimension (power of 2 in this casee)
- The thing is there is allot of correlation we can use in the channel

- Same concept applied to video signals, shown in Figure 1.5.



Figure 1.5: Video Compression

1.3 Steps of Compression

General steps of compression are shown in Figure 1.6.

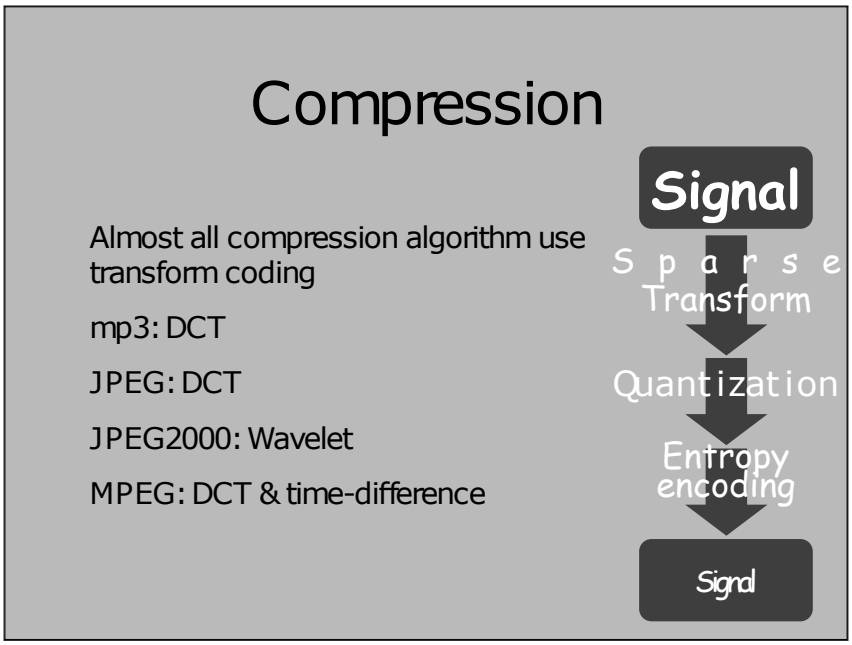


Figure 1.6: Compression Steps

- We apply some sparse transformation
  - Sparse transformation decompose the signals in some set of few coefficients
- We quantize these coefficients and do some entropy coding

1.4 Sparsity and Noise

Usually, when we have a sparse signal with noise, we can identify the signal directly by simple threshold operation. So in other words, sparse signals (or sparse representation in some other domains) have the ability to separate between the useful components of the signals (non zero component with large values usually) and with noisy components which tends to have small values. This is actually known as denoising by using sparsity. These concepts are illustrated in Figure 1.7.

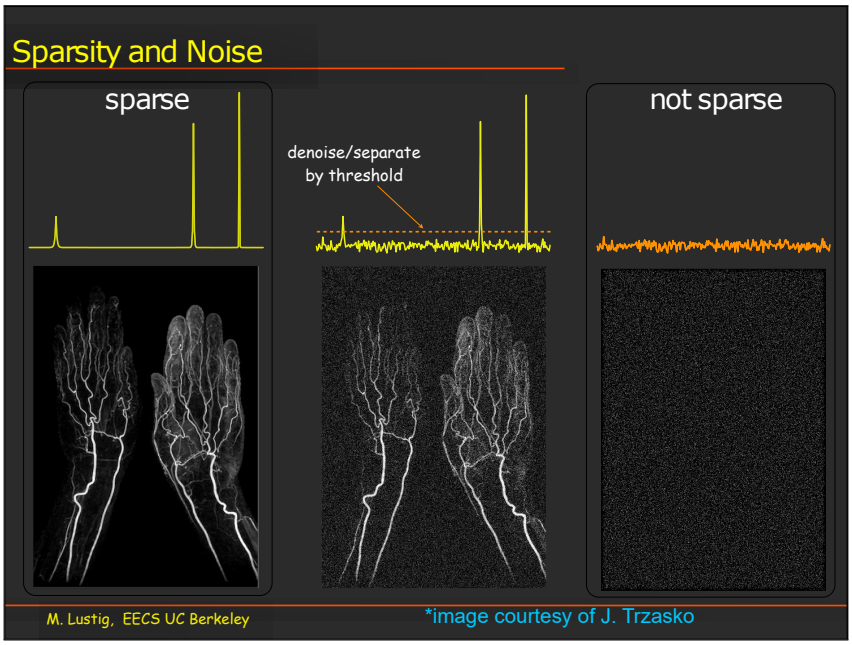


Figure 1.7: Noise Reduction taking advantage from sparse nature of signal

Some example of brain image using wavelet decomposition shown in Figure 1.8

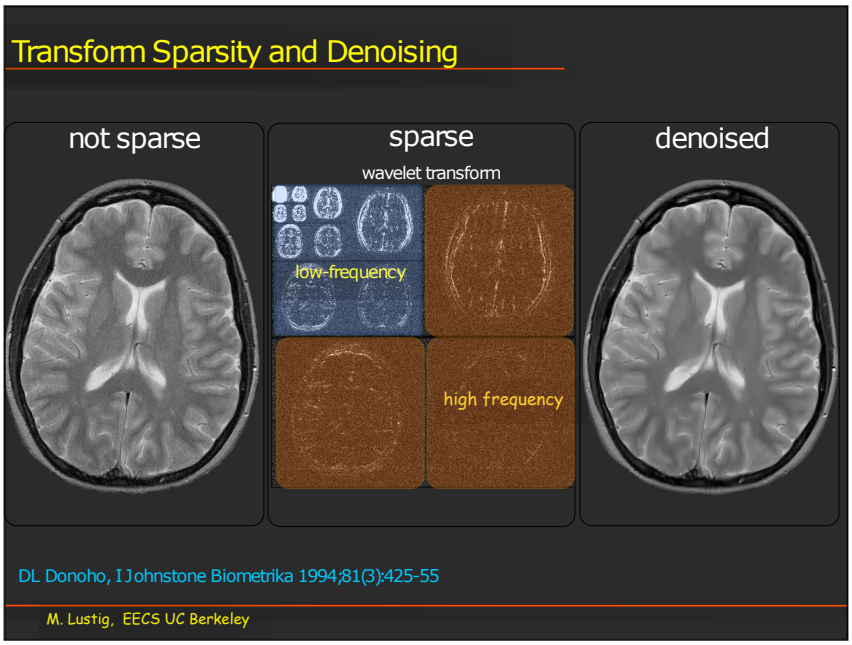


Figure 1.8: Noise Reduction using wavelet transform

- It is true that the image is not sparse. However its wavelet decomposition is sparse and hence we can use thresholding to reduce the noise level and reconstruct the image

1.5 Moving to Compressive Sampling

We have seen in sections before that sparsity help us with noise reduction and compression, but we didn't see its relation with compressive sampling, yet !.

Let's see the concepts of compressive sampling shown in Figure 1.9

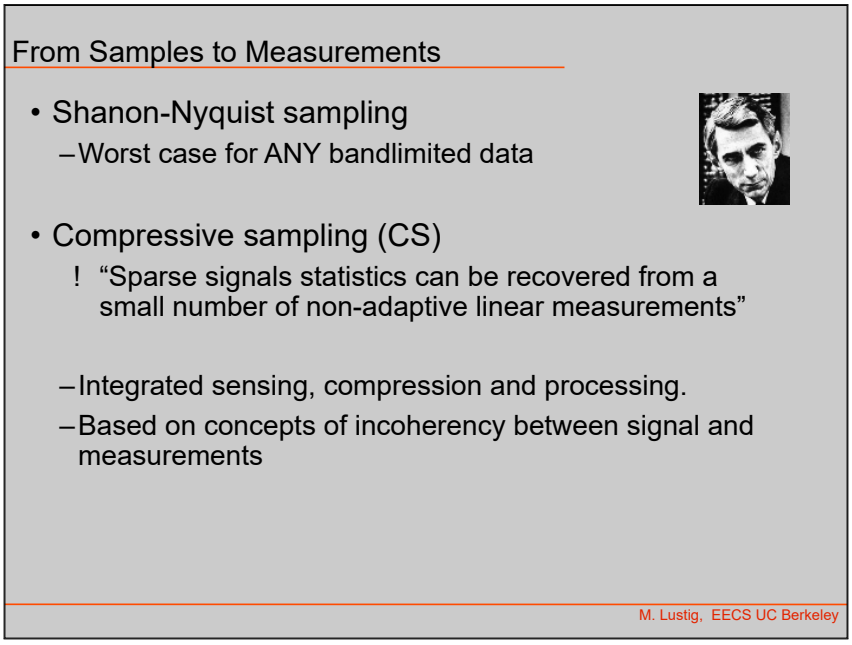


Figure 1.9: Compressive Sampling Concept

- Usually by following Nyquist theorem, we can reconstruct the full signal
- However in compressive sampling, we reduce the number of samples and still able to do reconstruction

Let's begin first by the ***classical sensing*** case, shown in Figure 1.10

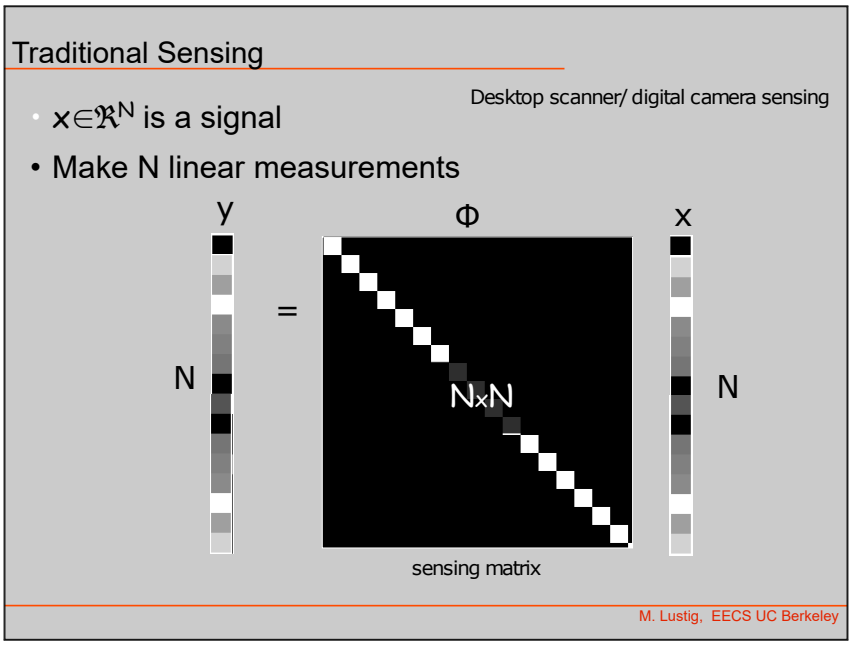


Figure 1.10: Classical Sensing Concept

- We have some signal $x \in \mathbb{R}^N$ and we need to recover it
- What we can do is to construct some N independent linear measurement vector y using a sensing matrix Φ
  - The matrix Φ shown in Figure 1.10 is an identity matrix related to single pixel camera experiment

- The best sensing matrix is the orthogonal matrix: $\Phi \times \Phi^T = \mathrm{I}$
  - Any orthogonal matrix can be used as a sensing matrix. Another example of sensing matrix Φ is shown in Figure 1.11, where it is the FFT matrix used in MRI

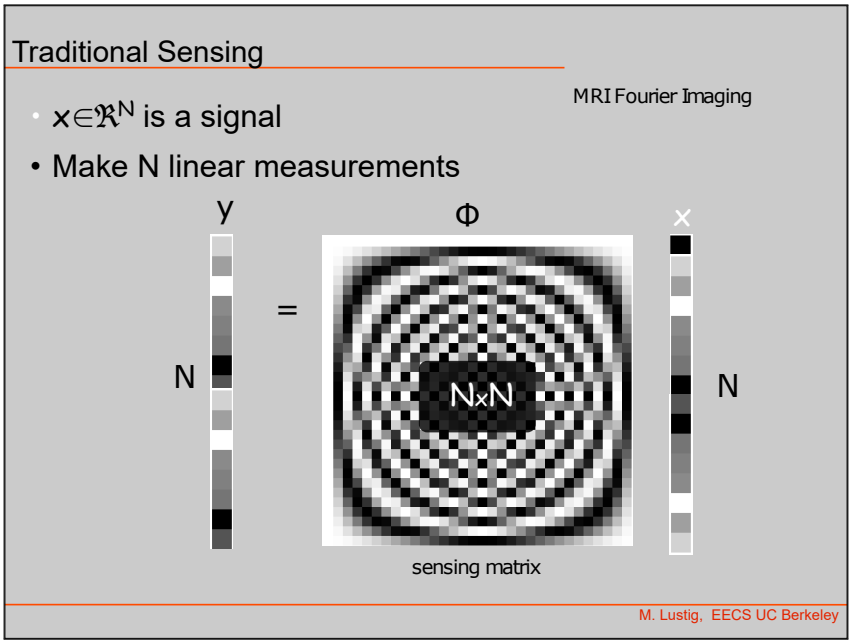


Figure 1.11: Classical Sensing: FFT matrix in MRI

So the classical case for estimating some unknown vector $x \in \mathbb{R}^N$, is to take the same number N on liner independent measurement.

Now let's shift the gears to the sparse case shown in Figure 1.12.

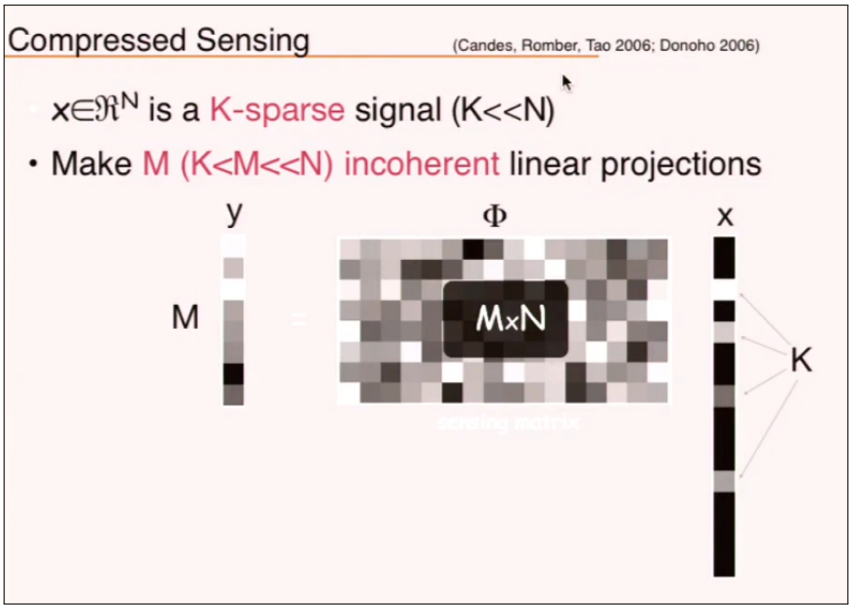


Figure 1.12: Sparse Sensing

- If my signal x is sparse, then we don't need to do N linear independent measurement
  - What we do is take M measurements such that M is far less than N, and little big than K, $\rightarrow$ $K < M << N$
- In the case of sparse also, the sensing matrix Φ becomes flat (rectangular), so it will not become fully orthogonal, but approximately orthogonal
  - What approximately orthogonal means that the columns of Φ will try to be orthogonal as much as they can
  - And also the product $\Phi \times \Phi^T$ will give an approximate identity matrix, where the elements off the diagonal tends to 0, but not exactly 0

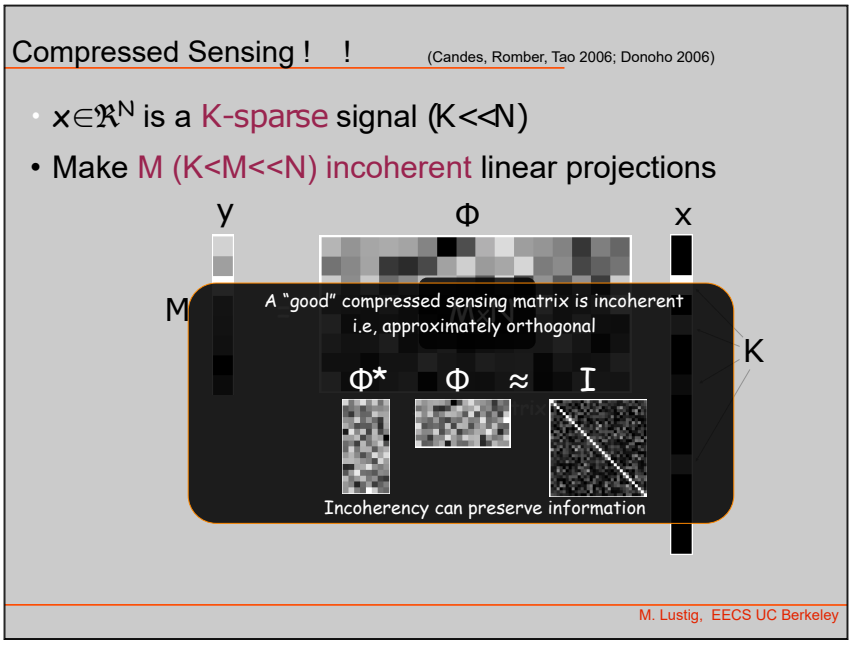


Figure 1.13: Sparse Sensing matrix Φ

If we look again at Figure 1.12, we can do the following:

- If we know where the non zero coefficient are in x, so we can recover x by eliminating the unnecessary columns in Φ which multiply the 0 elements
- In relation with band limited signals, we know where our frequency content is concentrated (the analogy with location of non zero components in signal x), so we can design some sampling criteria lower than the Nyquist rate in which we can recover the signal with no aliasing

However, we usually don't know where these non zero values are, so that's a problem.

However, we can benefit from the constraint that x is sparse, and try to formulate it in some optimization formulation. There are many ways to try:

- Some natural intuition is because x is sparse, we can try to minimize the energy subject to $y = \Phi x$, which can be written as:

$$\begin{aligned} &\min \parallel x \parallel_2 \\ &\text{s.t } y = \Phi x \end{aligned} \tag{1.1}$$

Chapter 2

Compressed Sensing 2: Dr Steve Brunton

2.1 References

Read chapter 3 in [1].

2.2 Notes

- A sparse representation: $x_{N\times 1} = \Psi_{N\times N} \times s_{N\times 1}$

  where $\begin{cases} s \text{ is a sparse vector} \\ x \text{ is the natural signal in the intial domain (time or spatial)} \end{cases}$

- Even if we compress and we are able to return x from x, the compresses sensing framework let us choose only few measurements and restore s.

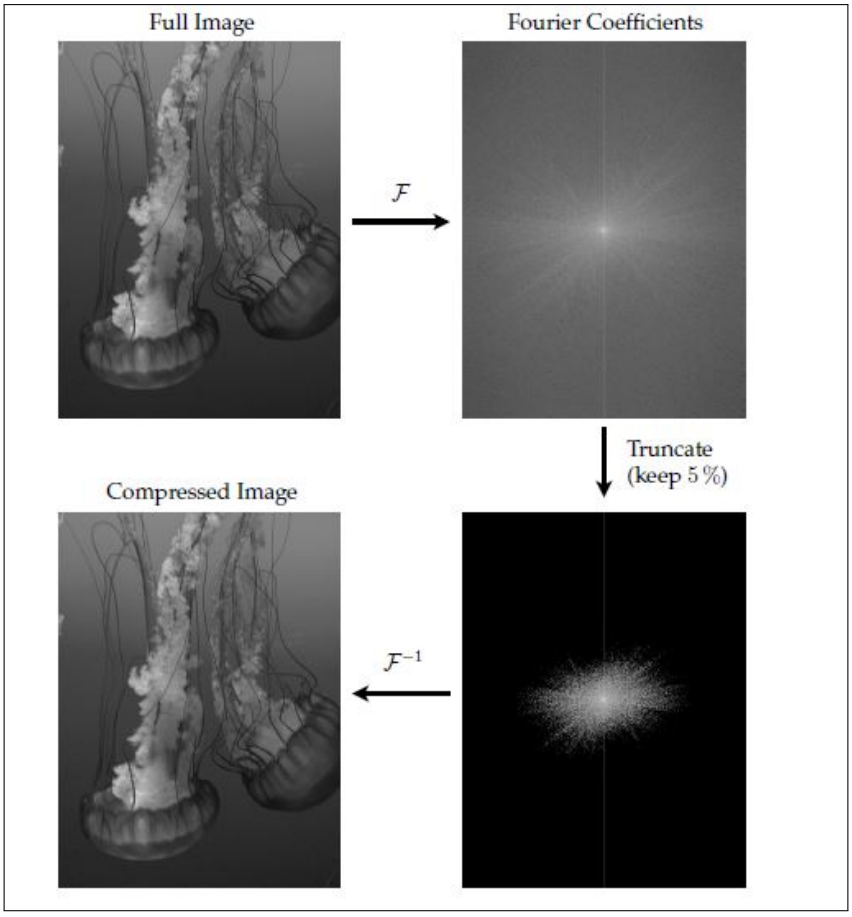


Figure 2.1: Compression using Fourier Basis

- However, in compression, we are making many measurements then throwing away most of these values
  - So what if we can directly make few measurements, recover the sparse vector s, then take the inverse Fourier to restore the original signal x
  - Mathematically, this can be done:

$$\begin{aligned} y_{m\times 1} &= C_{m\times N} \times x_{N\times 1} \\ &= C_{m\times N} \times \Psi_{N\times N} \times s_{N\times 1} \\ &= \Theta \times s_{N\times 1} \end{aligned} \tag{2.1}$$

- Mathematics:
  - The system of equation by (2.1) is known to be an undertimined system
    - There are infinite number of s vectors which satisfy equation (2.1)
  - However, we know that s should be sparse (a constraint which we need to satisfy), hence we add a regularization term to the objective function
    - This the work of many genius people in the field of optimization, statistics and programming
  - For many years, people have used the ℓ_2 norm

* ℓ_2 norm doesn't produce a sparse solution
* It minimize the radius of the circle (the energy) but all the component of the signal (vector) are still active

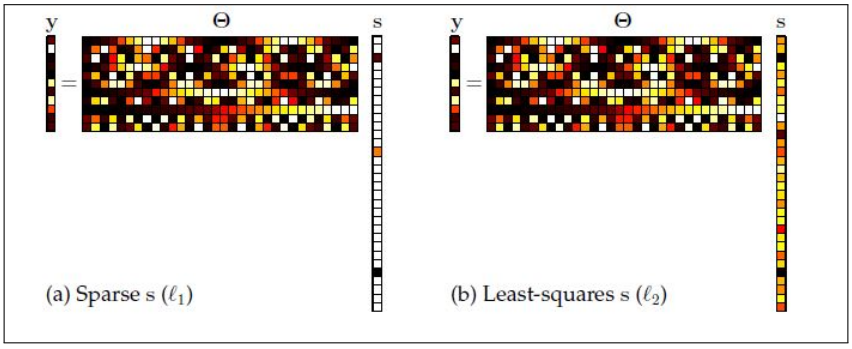


Figure 2.2: ℓ_1 and ℓ_2 norms effects

– Objective Function writing for this problem:

$$\min \| C_{m \times N} \times \Psi_{N \times N} \times s_{N \times 1} - y \|_2 + \lambda \|s\|_1 \tag{2.2}$$

Or we can formulate it as in (2.3):

$$\min \| s \|_1 \text{ subject to } \| C_{m \times N} \times \Psi_{N \times N} \times s_{N \times 1} - y \|_2 \leq \epsilon \tag{2.3}$$

where ϵ is a tolerance we set due to some noise in the measurement

2.3 Conditions

Recalling the matrix formulation:

$$\begin{aligned} y_{m \times 1} &= C_{m \times N} \times x_{N \times 1} \\ &= C_{m \times N} \times \Psi_{N \times N} \times s_{N \times 1} \\ &= \Theta \times s_{N \times 1} \end{aligned} \tag{2.4}$$

In order to solve the optimization problem, we need to be aware of several conditions to be respect:

- C is called the measurement matrix, where is has to satisfy some conditions

1. Incoherence with respect to Ψ: C needs not be parallel
2. Good measurement matrices can be Gaussian distributed, Bernoulli,$\cdots$.
   So they are random. Put picture here.
3. A bad measurement matrix example: if we take Ψ^T.
   If we do so, we won't capture all the frequencies (or features) of the sparse vector s. Put the picture here.
4. Note: it is usually that mathematicians love Gaussian distributed matrices because they are robust mathematically. However, this not usually true in engineering problems. For example, the single pixel camera measurement matrix is used often (to do this later)

- Number of measurements p : they have to be $\propto O\left(k \log\left(\frac{n}{k}\right)\right)$

Bibliography

[1] S. L. Brunton and J. N. Kutz, *Data-Driven Science and Engineering: Machine Learning, Dynamical Systems, and Control*, 1st ed. USA: Cambridge University Press, 2019.